

Aditya Kulkarni

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SUMMARY

MS CS candidate (May 2026) with production AI engineering experience across LLM product features, multi-model orchestration, and evaluation infrastructure. Built and shipped end-to-end AI features in TypeScript/Node.js/PostgreSQL -- Notion's exact stack -- while designing rigorous LLM eval frameworks with statistical testing. Passionate about making AI products more intelligent, reliable, and measurable at scale.

EXPERIENCE

- Product Engineer Intern

Isomer AI

May 2025 - Present

 - Shipped 10+ backend features across 7 production deployments in TypeScript/Node.js/Express/PostgreSQL/Redis; built multi-model LLM routing over 8+ models (Claude, GPT-4o, Gemini) via MCP with per-request latency and cost tracking.
 - Implemented Temporal.io durable orchestration for document processing pipelines, Auth0 RBAC, real-time push (Pusher), and Sentry observability; contributed 394 commits across the full product lifecycle.
- Graduate Research Assistant

NC State University

Apr 2025 - Present

 - Built cloud-native genomic data platform on Google Cloud Run serving 500 GB to 100+ researchers; improved ML pipeline accuracy by 12% via hyperparameter tuning.
- Software Developer Intern

Mphasis (HP)

Jun - Aug 2024

 - Delivered health-claim ML tool (+25% accuracy via OCR/NLP); automated AWS/GCP CI/CD and added Grafana monitoring dashboards.
- AI/ML Intern

Capgemini

Jun - Aug 2023

 - Built Hadoop+Spark batch pipelines and Kafka real-time ingestion layer; reduced processing time 15% and latency 20%.

PROJECTS

- LLM Eval Harness

Evals / Benchmarking

Vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, Gemini 2.5 Pro on 2,000 frozen examples (grounded QA, multi-hop QA, FEVER). Tracked hallucination rate, TokenF1, p50/p95 latency, cost/call with paired bootstrap CIs, McNemar test, Holm correction. DuckDB + Parquet + Streamlit dashboard.

Python · DuckDB · Parquet · Streamlit · PyTorch · Hugging Face
- Glance AI

RAG / AI Product

Compliance-ready RAG system for fintech/healthcare with per-tenant data isolation at vector retrieval layer; zero cross-tenant context leakage by design; deterministic output architecture.

Python · LangChain · RAG · FastAPI · PostgreSQL
- AlgoPulse

LLM Orchestration

Re-architected from API polling to event-driven streaming; Temporal.io durable workflow state + Kafka partitioned workers; 2,000 async LLM inference req/sec, zero message loss.

TypeScript · Temporal.io · Kafka · Supabase Edge Functions
- VectorLift

Neural IR

BM25 + dense bi-encoder + cross-encoder re-ranker on MS MARCO; Kafka-driven indexing into Elasticsearch + Qdrant; FastAPI serving; NDCG@10/MRR@10 evaluation; Prometheus + Grafana.

Python · Elasticsearch · Qdrant · Kafka · FastAPI · Prometheus

EDUCATION

- MS Computer Science — NC State University

Aug 2024 - May 2026

AI, OS, Data Science, Software Engineering, Algorithms
- BTech Computer Science — Mumbai University

Aug 2020 - May 2024

CGPA 9.32 / 10

SKILLS & AWARDS

- Languages:** TypeScript, Python, JavaScript, Java, SQL, C **Frameworks:** Node.js, Express, React, FastAPI, Spring Boot, LangChain
- AI/ML:** LLMs, RAG, MCP, LLM Evals, PyTorch, Hugging Face, Embeddings, Prompt Engineering
- Infra:** PostgreSQL, Redis, MongoDB, Kafka, Docker, Kubernetes, AWS, GCP, GitHub Actions, Prometheus, Grafana
- HackNC24 Top 10** / 100+ teams | **IEEE Published** (Mapify, 500+ attendees) | **Sanjeevani Top 1%** / 1,100+ submissions | **Google DSC Lead** (500+ students)